

Math

– 22 Questions | 35 minutes –

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

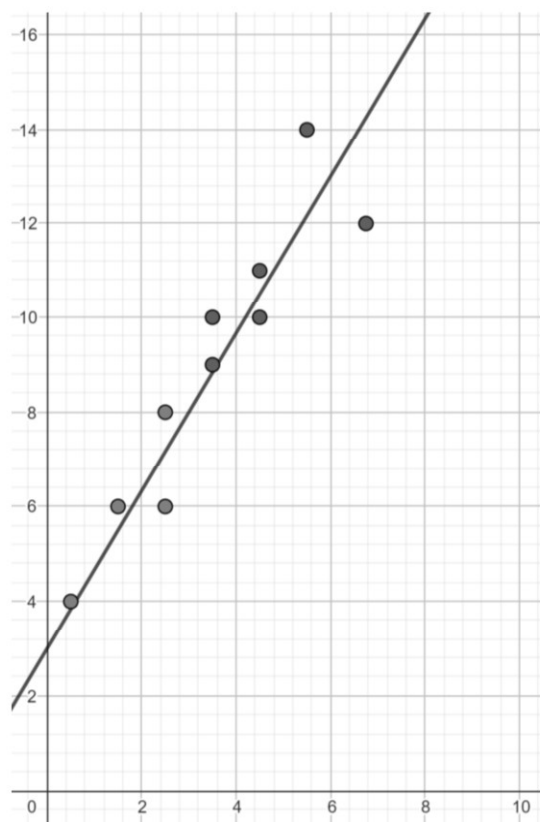
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1

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A** $y = \frac{5}{3}x + 3$
- B** $y = -2x + 1.5$
- C** $y = 3x + 1.5$
- D** $y = -3x + 1.5$

2

$$3x - 4y = 5$$

$$x = 7$$

Which is the solution (x, y) to the given systems of equations?

- A** $(7, 4)$
- B** $(7, 12)$
- C** $(7, -12)$
- D** $(7, -4)$

3

What value of x is the solution to the equation $20x + 24 = 28x$?

- A** $\frac{1}{4}$
- B** $\frac{1}{3}$
- C** 3
- D** 5

4

The function f is defined by $f(x) = \frac{7}{10}x + 74$. What is the value of $f(20)$?

5

The function f is defined by $f(x) = 8 + \sqrt{x}$. What is the value of $f(36)$?

- A 0
- B 14
- C 17
- D 26

6

Triangles ABC and DEF are congruent, where A corresponds to D and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

7

At a convention center, there are a total of 325 visitors. Each visitor is located in either room A , room B or room C . If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.16. How many visitors are located in room C ?

- A 25
- B 36
- C 104
- D 117

8

$$g(m) = -0.05m + 14.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

- A 0.05
- B 14.1
- C 20
- D 282

9

What is the center of the circle in the xy -plane defined by the equation $(x - 9)^2 + (y + 6)^2 = 81$?

- A $(-9, -6)$
- B $(-9, 6)$
- C $(9, -6)$
- D $(9, 6)$

10

If $2x + 3 = 9$, what is the value of $4x - 4$?

11

$$f(x) = 6,000(0.7)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 6,000 in this context?

- A The estimated percent decrease in the insect population for this species and area every 8 years after 2008
- B The estimated percent decrease in the insect population for this species and area each year after 2008
- C The estimated insect population for this species and area 8 years after 2008
- D The estimated initial insect population for this species and area in 2008

12

Line r is defined by the equation $4x - 5y = 8$. Line s is parallel to line r in the xy -plane. What is the slope of line s ?

- A $\frac{5}{4}$
- B $\frac{4}{5}$
- C -4
- D -5

13

The cost to rent a bus from Company X is \$750 for the first 2 hours and an additional \$50 per hour for each hour after the first 2 hours. If the total cost to rent the bus from Company X for t hours, where $t > 2$, is \$1,050, which equation represents this situation?

- A $750(t - 2) + 50t = 1,050$
- B $750(2t) + 50t = 1,050$
- C $750 + 50(t - 2) = 1,050$
- D $750 + 50(2t) = 1,050$

14

$$y = -\frac{1}{4}x^2 + x + 28$$

The given equation models a company's active projects over 6 months, where y is the estimated number of active projects x months after the end of April 2012, where $0 < x < 6$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A At the end of April 2012, the estimated number of active projects was 0.
- B At the end of April 2012, the estimated number of active projects was 28.
- C At the end of May 2012, the estimated number of active projects was 0.
- D At the end of May 2012, the estimated number of active projects was 28.

15

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{4\sqrt{2}}{9}$. What is the value of $\cos(S)$?

- A $\frac{4\sqrt{2}}{9}$
- B $\frac{4\sqrt{2}}{7}$
- C $\frac{7\sqrt{2}}{8}$
- D $\frac{9\sqrt{2}}{8}$

16

$$19.5x + 29.75y = 394$$

Odalys ordered mulch and river rock, which cost a total of \$394, for her home. The given equation represents the relationship between the number of cubic yards of mulch, x , and the number of tons of river rock, y Odalys ordered. How much more, in dollars, did a ton of river rock cost Odalys than a cubic yard of mulch? Enter decimal form only.

17

A hemisphere is half of a sphere. If a hemisphere has a radius of 50 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A 5,200
- B 20,900
- C 196,300
- D 261,800

18

In triangle ABC , angle B is a right angle, and the length of BC is 248 millimeters. If $\cos(A) = \frac{3}{5}$, what is the length, in millimeters, of AB ?

- A 62
- B 186
- C 248
- D 310

19

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5% of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.97 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

- A $0.965(6.97)^{10}$
- B $1.035(6.97)^{10}$
- C $6.97(0.035)^{10}$
- D $6.97(0.965)^{10}$

20

To study fluctuations in leaf water potential, samples of wood were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 2.72 grams, what is the density of this cube, in grams per cubic centimeter? Enter decimal form only.

21

x	y
-11	-25
9	55

The table shows two values of x and their corresponding values of y . The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{3}, a\right)$. What is the value of a ? Enter decimal form only and round to two decimal places.

22

Which system of linear equations has no solution?

- A $-2x + 8y = -5$
 $2x - 8y = 5$
- B $2x - 8y = 5$
 $3x + 9y = 6$
- C $2x - 8y = 5$
 $-6x + 24y = -15$
- D $-2x + 8y = 5$
 $4x - 16y = 10$

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

Math

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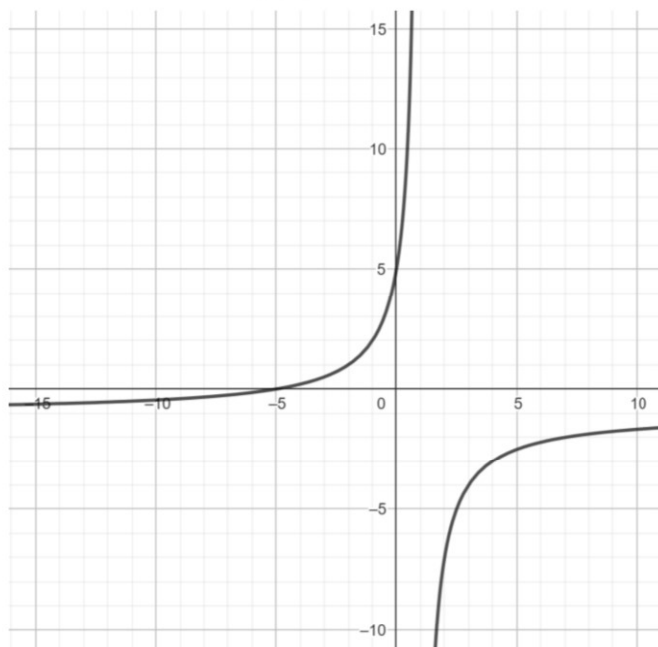
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1

The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0)$?



2

Which expression is equivalent to $(9nr^2 + 3nr) + (6n^2r + 5nr)$?

- A $15n^2r^2 + 8nr$
- B $15n^3r^3 + 8n^2r^2$
- C $54n^3r^3 + 15n^2r^2$
- D $9nr^2 + 6n^2r + 8nr$

3

If $\frac{2a}{b} = 5.8$ and $\frac{a}{bn} = 23.2$, what is the value of $\frac{1}{n}$?

4

$$v = -\frac{w}{161x}$$

The given equation relates the distinct positive numbers v , w and x . Which equation correctly expresses w in terms of v and x ?

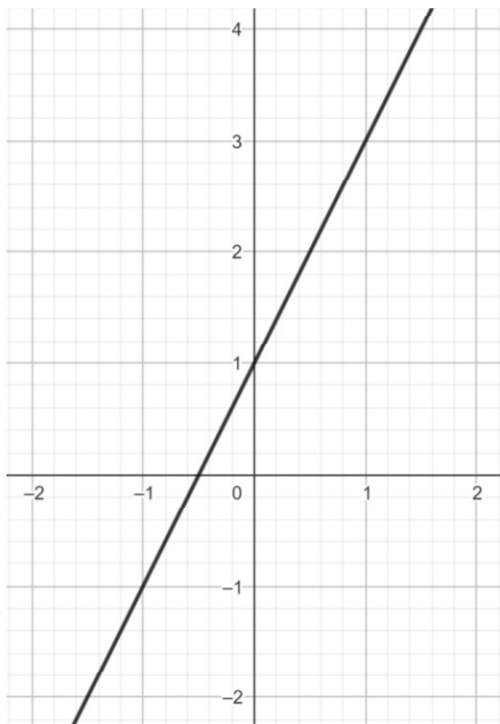
- A $w = -161vx$
- B $w = -\frac{161v}{x}$
- C $w = -\frac{x}{161v}$
- D $w = v + 161x$

5

A real estate company offers a series of three webinars. 1,250 people attended the first webinar. 68 of the people who attended the first webinar attended the second webinar, and 32 of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?

6

The graph of $y = f(x)$ is shown. What is the y -intercept of the graph of $y = f(x) + 13$?



- A $(0, -12)$
- B $(0, 14)$
- C $(1, 13)$
- D $(1, 14)$

7

The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(7) - f(a) = -15$?

- A -22
- B 3
- C 10
- D 75

8

If $-9(15 - 3x) + 5 = -10(15 - 3x) + 21$, what is the value of $15 - 3x$?

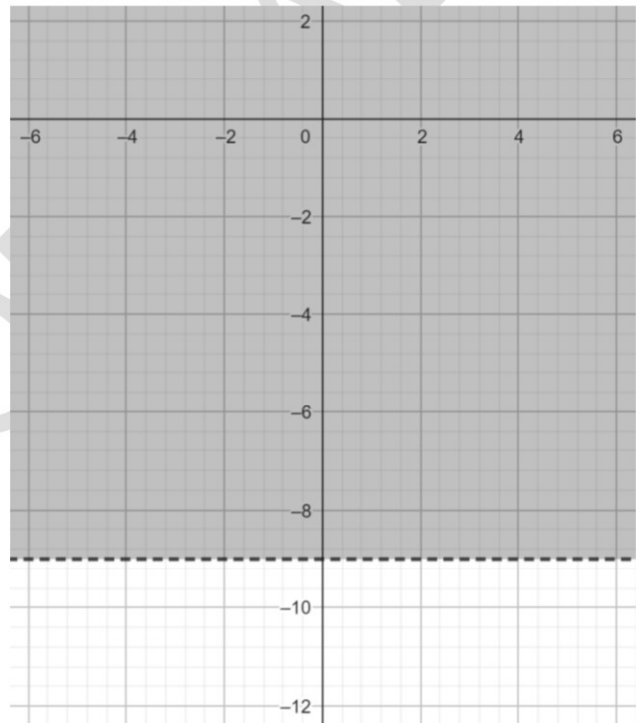
9

The minimum value of x is 11 less than 8 times another number n . Which inequality shows the possible values of x ?

- A $x \leq 8n - 11$
- B $x \geq 8n - 11$
- C $x \leq 11 - 8n$
- D $x \geq 11 - 8n$

10

The shaded region shown represents the solutions to the inequality $-18y < c$, where c is a constant. What is the value of c ?



- A 162
- B 9
- C -9
- D -162

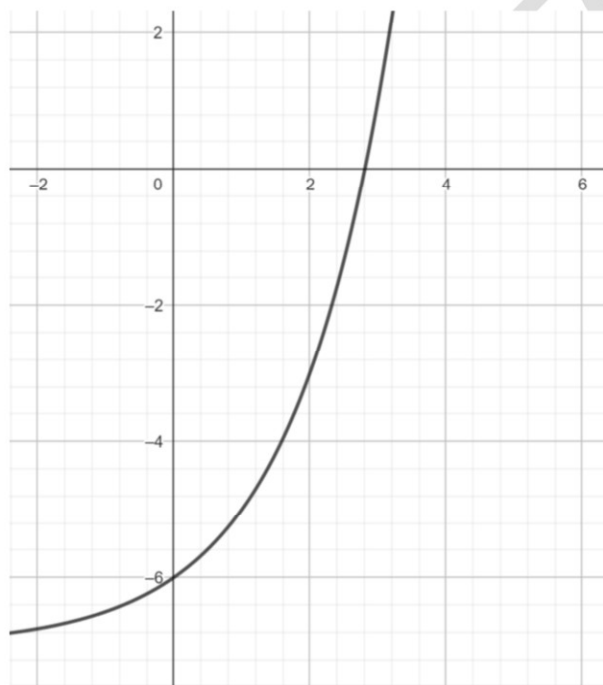
11

At a constant temperature, a scuba diver combines two different concentrations of nitrox mix (oxygen-nitrogen gas mixture) to fill a tank to a total pressure of 130 atmospheres (*atm*), of which 34 is contributed by oxygen. Nitrox mix *A* contributes x *atm* of pressure to the tank, of which 22 is contributed by oxygen. Nitrox mix *B* contributes y *atm* of pressure to the tank, of which 48 is contributed by oxygen. Which system of equations represents this situation?

- A $x + y = 130$
 $0.48x + 0.22y = 0.34(130)$
- B $x + y = 130$
 $0.22x + 0.48y = 0.34(130)$
- C $x + y = 0.34(130)$
 $0.48x + 0.22y = 130$
- D $x + y = 0.34(130)$
 $0.22x + 0.48y = 130$

12

The graph of the equation $y = 2^x + k$ is shown, where k is a constant. What is the value of k ?



- A -7
- B -6
- C -5
- D -4

13

A certain town has an area of 3.99 square miles. What is the area, in square yards, of this town (1 mile = 1,760 yards)?

- A 441
- B 7,022
- C 776,341
- D 12,359,424

14

The function f is defined by $f(x) = 4x^2$. What is the value of $f(11)$?

- A 44
- B 52
- C 88
- D 484

15

The function f gives the product of a number, x , and a number that is 66 more than x . Which equation defines f ?

- A $f(x) = x^2 + x + 66$
- B $f(x) = x^2 + 66$
- C $f(x) = x^2 + 66x$
- D $f(x) = x^2 + 66x + 66$

16

One of the factors of $4x^3 + 88x^2 + 468x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

17

Alex engages in up to 3 types of exercise each week for a total of 9 hours while training for a triathlon. Alex swims for the same number of minutes each week. The equation $y = 540 - x - 200$ represents the situation where Alex bikes for x minutes during a week and runs for any remaining training time y , in minutes. If this equation is graphed in the xy -plane, which statement is the best interpretation of the x -intercept of the graph?

- A During a week when Alex swims for 340 minutes, he bikes for 200 minutes.
- B Each week, Alex bikes and runs for a total of 340 minutes.
- C During a week when Alex doesn't bike, he runs for 340 minutes.
- D During a week when Alex doesn't run, he bikes for 340 minutes.

18

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 190)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A $(-18x + 380)^\circ$
- B $(-18x + 740)^\circ$
- C $(18x - 380)^\circ$
- D 180°

19

$$\frac{4}{7}(4x + 7)\left(x + \sqrt{4k + 7}\right)\left(x - \sqrt{4k + 7}\right) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 77. What is the value of k ? Enter decimal form only.

20

At the time of posting a video, a social media channel had 100 subscribers. Each day for five days after the video was posted, the number of subscribers tripled from the number the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the video was posted?

- A $n = (100)^d$
- B $n = 100(3)^d$
- C $n = 100\left(\frac{1}{3}\right)^d$
- D $n = (100)^3 + d$

21

	Volume (cubic units)
Right circular cylinder A	112π
Right circular cylinder B	$3,024\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.

22

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(-8, 1)$, contains the point $(-9, -1)$, and has a y -intercept at $(0, a)$. The graph of $y = 6f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

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